ICT162

Timed Online Assignment – January Semester 2022

Object Oriented Programming

Thursday, 26 May 2022 **4:00 pm – 6:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FOUR (4)** questions and comprises **TWELVE (12)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, Your Name, and SUSS PI No. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAICT162_RaphaelLee_T1923161 (omit D/O, S/O).*

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the Word file provided it does not exceed the file size limit of 500MB. Images of handwritten answers will not be marked.

7. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

8. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

Answer all questions. (Total 100 marks)

Read the following introduction before attempting the questions.

In the game of Golf, golfers have a collection of tools that they used to get the golf ball from the tee box to the hole. These tools are called golf clubs. A golfer is allowed to carry these clubs in his golf bag and free to decide what the final makeup ultimately looks like. The diagram in Figure A shows the 3 categories of golf clubs.

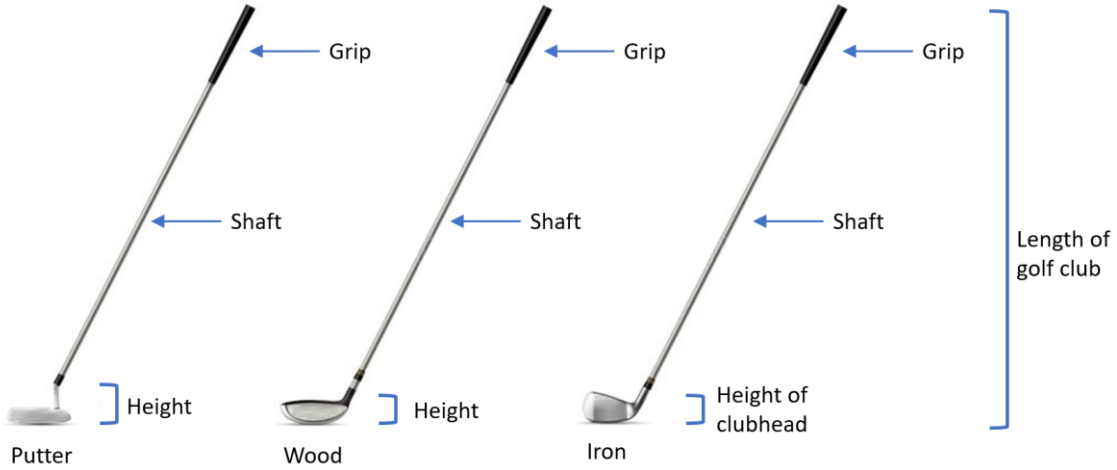


Figure A

Each club is made up of:

- Clubhead: This part of the club is located at the bottom.
  - All heads have loft and weight. Loft is the angle created between the face of the clubhead and the ground and will be different for every club. The loft and how fast the clubhead is swung will have a direct impact on the distance the ball travels.
  - The following are the 3 common types of clubhead:

Putter	Wood	Iron
Putter heads has different styles like blade, half-mallet, and mallet. The putter is used to roll the ball forward into the hole.	Woods are bigger and rounder clubheads that are designed to shoot the ball over long distances. The size of the head is measured in cubic centimetre (cc).	Made of hard (cast) or soft (forged) iron. Irons are the most represented clubs in a golfer's bag –numbering from 3-iron to 9-iron, pitching and sand wedge.

- Grip: Top part of the club, where golfers will hold. Can be rubber, leather or synthetic.
- Shaft: Middle part, either graphite or steel, with different flex. Bottom end of the shaft is connected to the clubhead, and the top part is installed with the grip.

Question 1

Study the class diagram in Figure Q1, which focuses on the clubhead class and its subclasses.

Demonstrate your understanding of how class hierarchy and association are used to organize information, and then construct the class hierarchy and association according to specification.

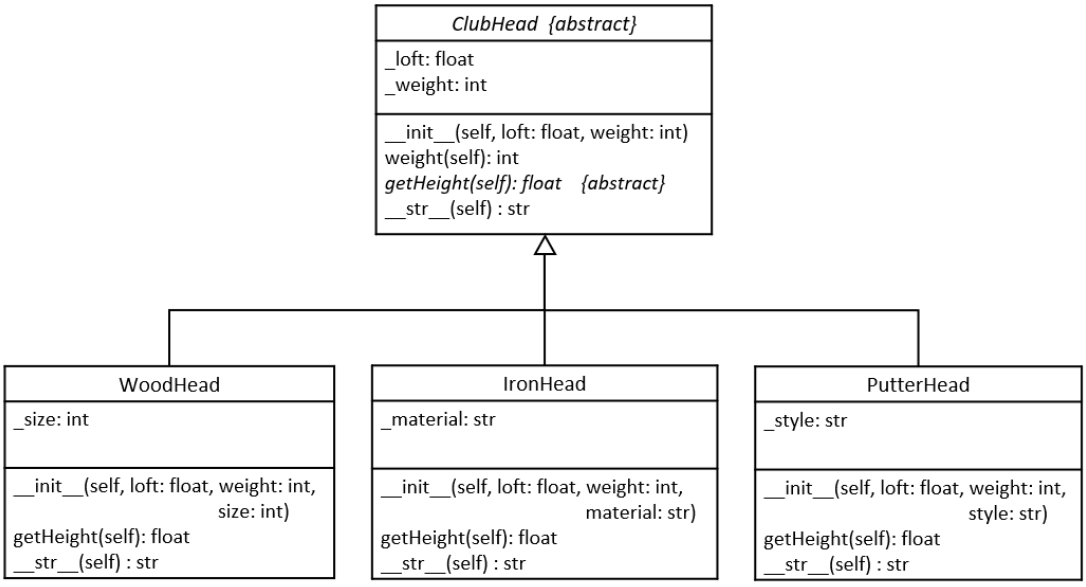


Figure Q1

(a) Implement the abstract class – ClubHead.

The ClubHead class has:

- Two instance variables:
  - _loft (float): loft of the clubhead in degrees.
  - _weight (int): weight of the clubhead in grams.
- Constructor initialises these 2 instance variables: _ loft and _ weight.
- Getter method for the instance variable: _weight. Use the property decorator.
- One abstract method, getHeight which returns the height of the clubhead.
- The __str__ method returns a string representation of a ClubHead object, which should include the loft and weight. Here are 3 examples in the suggested format:

```
Loft:  9.0 degrees      Weight: 200g
Loft: 37.0 degrees      Weight: 250g
Loft:  3.0 degrees      Weight: 380g
```

(6 marks)

(b) Write the Python class definitions for the WoodHead, IronHead and PutterHead.

The WoodHead class is a subclass of ClubHead, and it has:

- One additional instance variable: _size (int) representing the head size in cubic centimetre (cc).
- Implement the method getHeight that returns the height of the WoodHead. The height is calculated by dividing the size by 400. Hence, for head size 400cc, the height is 1 inch, while head size of 200cc is 0.5 inch.
- The __str__ method returns a string representation of a WoodHead object. This is an example in the suggested format:

```
Loft:  9.0 degrees  Weight: 200g      Head Size: 330cc
```

The IronHead class is also a subclass of ClubHead.

- One additional instance variable: _material (str) indicating if the head is made using cast or forged iron.
- For IronHead, the method getHeight returns 1 inch as iron design is limited.
- The __str__ method returns a string representation of a IronHead object. Below are two examples in the suggested format:

```
Loft: 37.0 degrees  Weight: 250g      Material: Cast
Loft: 56.0 degrees  Weight: 280g      Material: Forged
```

The PutterHead class is also a subclass of ClubHead.

- One additional instance variable: _style (str) indicating if the putter is a “Blade”, “Half-Mallet” or “Mallet”.
- For PutterHead, the method getHeight returns 1 inch if the putter is a “Blade”. Otherwise, the method returns 0.5 inch.
- The __str__ method returns a string representation of a PutterHead object. Below are two examples in the suggested format:

```
Loft:  3.0 degrees  Weight: 380g      Style: Blade
Loft:  3.5 degrees  Weight: 360g      Style: Mallet
```

(14 marks)

(c) Write Python statements for the following:

(i) Create the appropriate ClubHead objects using the information below.

ClubHead	Loft (degrees)	Weight (grams)	Others
Putter	3.5	365	Style: Blade
Iron	37.5	285	Material: Forged
Wood	9.5	206	Size: 450cc

(ii) Print the details and height of the created ClubHead objects in Q1(c)(i).

(5 marks)

Question 2

As shown in Figure A, the golf club has 3 parts – the clubhead, shaft and grip. In Figure Q2, the class diagram is updated with Shaft, Grip and Club classes, to mimic the construction of golf club.

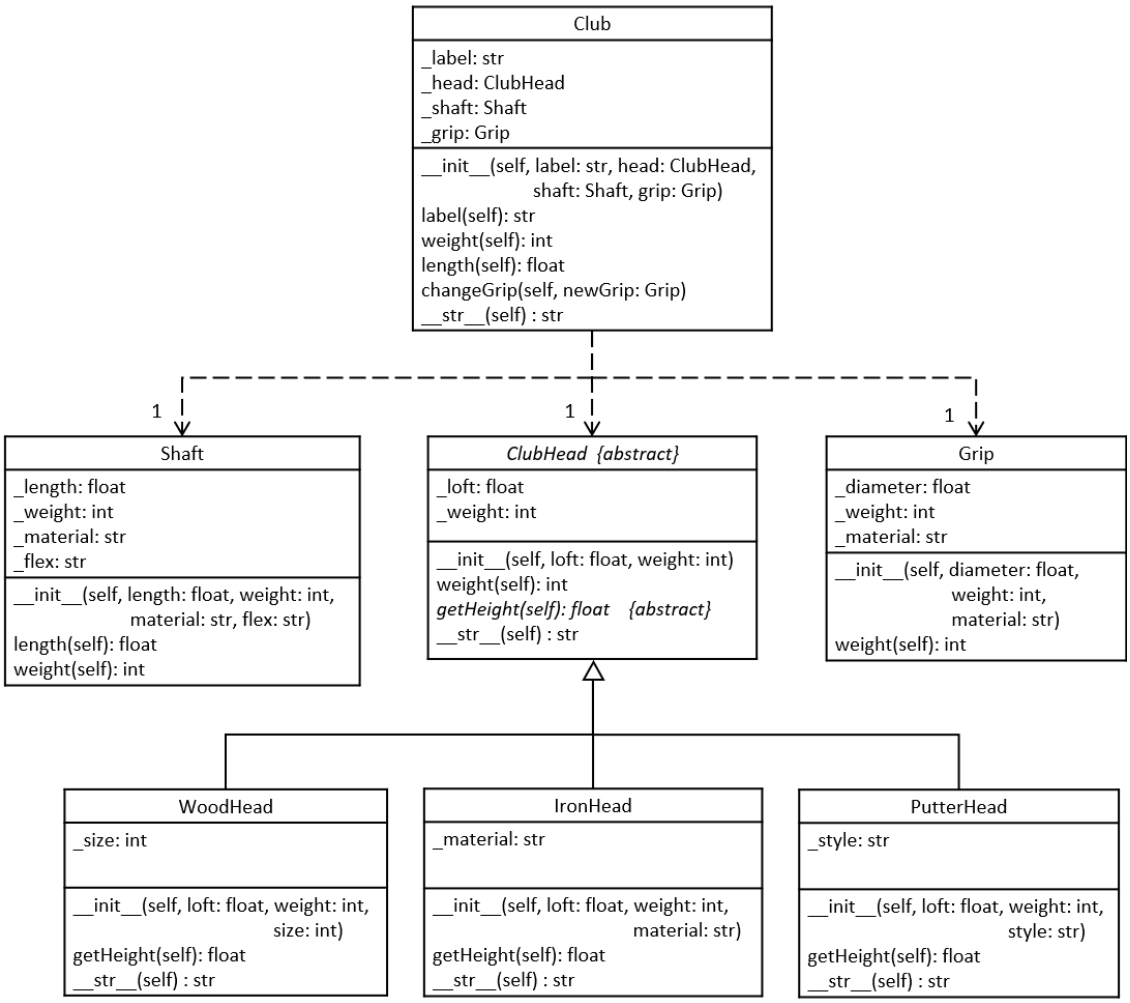


Figure Q2

Assume that the python code for Shaft and Grip classes are already provided.

(a) Construct and Implement the Club class.

The Club class has:

- Four instance variables:
  - _label (str): labelling or numbering to help golfers identify the club. For example, “7-iron”, “3-wood” and “4-hybrid”.
  - _head (ClubHead): representing the clubhead object for this club.
  - _shaft (Shaft): representing the shaft for this club.
  - _grip (Grip): the grip that is installed for this club (onto the top end of the shaft).
- Constructor initialises the 4 instance variables using the given parameters.
- Getter method for instance variable label. Use the property decorator.

- Getter methods for length and weight. Use the property decorator
    - Length of the club is the length of the shaft and the height of the clubhead.
    - Weight of the club is the total weight of clubhead, shaft and grip.
- The changeGrip method replaces the instance variable _grip with the given parameter Grip object.
- The __str__ method returns a string representation of a Club object. Below are two examples in the suggested format:

```
Club: 3-wood      Length: 43.43in        Weight: 348g

Club: 7-iron      Length: 37.00in        Weight: 434g
```

(9 marks)

(b) Write Python statements to perform the following:

(i) Create **THREE (3)** golf clubs using the information below.

Label	ClubHead	Loft (degrees)	Weight (grams)	Others
Driver	Wood	10.5	203	Size: 450cc
8-Iron	Iron	34.5	268	Material: Cast
Sunset	Putter	3	380	Style: Mallet

Label	Shaft Flex	Shaft Length (inches)	Shaft Weight (grams)	Shaft Material
Driver	Stiff	45	65	Graphite
8-Iron	Regular	35.5	109	Steel
Sunset	Stiff	33	120	Steel

All grips are 0.6 inches in diameter, 62g in weight and made of "Rubber".

(ii) Print the total weight of the 3 clubs created in Q2(b)(i).

(iii) Replace the grips for all the golf clubs as they are worn out. The new grips are made of "Leather", 65g in weight and 0.58 inches in diameter.

(iv) Describe the terminology polymorphism and provide an example using your python statements in Q2(b) that exhibits polymorphism. Justify your answer.

(10 marks)

(c) Differentiate between object composition and inheritance. Referencing Figure Q2, identify the relationship between the following pairs of classes and justify your choices:

(i) Club and Shaft

(ii) ClubHead and PutterHead

(iii) Club and IronHead

(6 marks)

Question 3

A golf bag is used to carry the golf clubs during the golf game. Study the class diagram in Figure Q3, which now feature the GolfBag class.

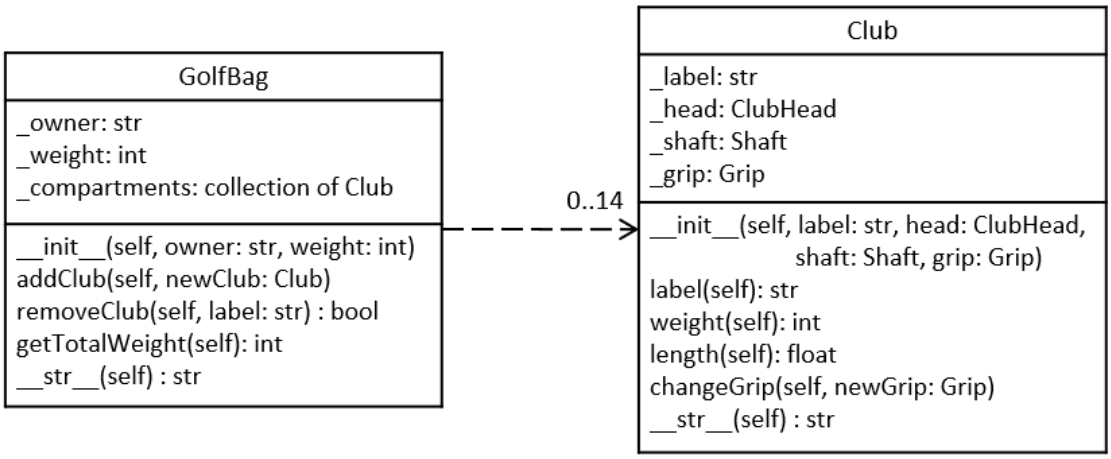


Figure Q3

(a) Implement the GolfBag class.

- There are three instance variables:
  - _owner (str): the golf bag owner's name.
  - _weight (int): weight of the golf bag when empty (no clubs), in grams.
  - _compartments (collection): you can decide to implement either Dictionary or List for collection of Club objects.
- Constructor initialises these 2 variables: _owner and _weight, using the given parameters, and set _compartments to an empty collection.
- The addClub method has a Club object as parameter and will add this new Club object into the _compartments collection.
- Given a club's label (str) as parameter, the removeClub method will search and remove the first occurrence of the matching Club object in the collection. The method returns True if the removal is successful, and False otherwise.
- Method getTotalWeight returns the sum of all clubs' weight inside the _compartments and the weight of bag.

- The __str__ method returns a string representation of a GolfBag object, which include the owner, all the clubs' label, length, weight and finally the total weight.

Here is an example in the suggested format:

```
Golf Bag Owner: Tan B Y
Club: Driver      Length: 46.12in        Weight: 333g
Club: 3-hybrid    Length: 40.38in        Weight: 385g
Club: 4-iron      Length: 38.50in        Weight: 417g
Club: 5-iron      Length: 38.00in        Weight: 417g
Club: 6-iron      Length: 37.50in        Weight: 417g
Club: 7-iron      Length: 37.00in        Weight: 417g
Club: 8-iron      Length: 36.50in        Weight: 417g
Club: 9-iron      Length: 36.00in        Weight: 417g
Club: P-Wedge     Length: 35.50in        Weight: 417g
Club: S-Wedge     Length: 35.25in        Weight: 417g
Club: Putter      Length: 34.00in        Weight: 540g
Total weight: 6,794g
```

(11 marks)

(b) State at least **TWO (2)** reasons to justify your choice of collection for the implementation of GolfBag's compartments.

(4 marks)

(c) The R&A (Royal and Ancient Golf Club of St Andrews) and the USGA (United States Golf Association) has recently update the equipment rules:

- The maximum number of clubs in a bag is 14. However, it is not necessary that the 14 clubs are unique. Hence, a golfer is allowed to carry TWO "7-iron" in his/her golf bag.
- The maximum weight of the golf bag, including the golf clubs is 7500g.

Table Q3

Write/add python codes to handle the new equipment rules:

(i) Define the class GolfEquipmentException, which is a subclass of the Exception class. This class has no additional attribute or method.

(ii) The new equipment rules in Table Q3 affect your class implementation for GolfBag. Modify the GolfBag class to raise GolfEquipmentException when any of the rules in Table Q3 is violated.

Furthermore, ensure your added/modified codes can easily cater to R&A and USGA changes in the equipment rules. For example, there may be new rules in future to increase the maximum number of clubs to 15, or maximum weight is to 9500g.

Submit your codes for Q3(a) separately from Q3(c).

(10 marks)

Question 4

A Golf Distance Calculator GUI is developed to allow golfers to check how far they can hit the golf ball when they provide their swing speed in mph. The GUI design of the Golf Distance Calculator is shown in Figure Q4.1.

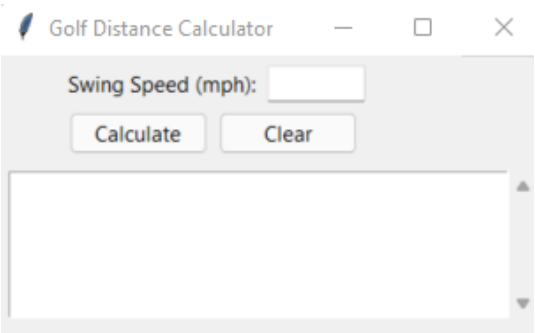


Figure Q4.1

You are given partial code in Table Q4 that produces the GUI in Figure Q4.2

```
import tkinter as tk
from tkinter import ttk, scrolledtext

class GolfDistanceCalculatorGUI:
    def __init__(self):
        self._win = tk.Tk()
        self._win.title("Golf Distance Calculator")
        self._win.resizable(True, True)
        self._win.geometry("320x180")      # resolution may differ
        self.create_widgets()
        self._win.mainloop()

    def create_widgets(self):
        dataFrame = ttk.Frame(self._win)
        dataFrame.grid(column=0, row=0, padx=2, pady=4)

        swingSpeed_lbl = ttk.Label(dataFrame, text="Swing Speed (mph):")
        swingSpeed_lbl.grid(column=0, row=0, padx=4, pady=4)

        actionFrame = ttk.Frame(dataFrame)
        actionFrame.grid(column=0, row=1, columnspan=2)
        self._calculate_btn = ttk.Button(actionFrame, text="Calculate")
        self._calculate_btn.pack(side = tk.LEFT, padx=4, pady=4)
        self._clear_btn = ttk.Button(actionFrame, text="Clear")
        self._clear_btn.pack(side = tk.LEFT, padx=4, pady=4)

        outputFrame = ttk.Frame(self._win)
        outputFrame.grid(column=0, row=1, padx=8, pady=4, columnspan=2)
        self._scrol_stxt = scrolledtext.ScrolledText(outputFrame, width=35,
                                                     height=5, wrap=tk.WORD)
```

```
        self._scrol_stxt.grid(column=0, row=0)
        # disabled to disallow data entry into scrolledText
        self._scrol_stxt.config(state=tk.DISABLED)
```

Table Q4

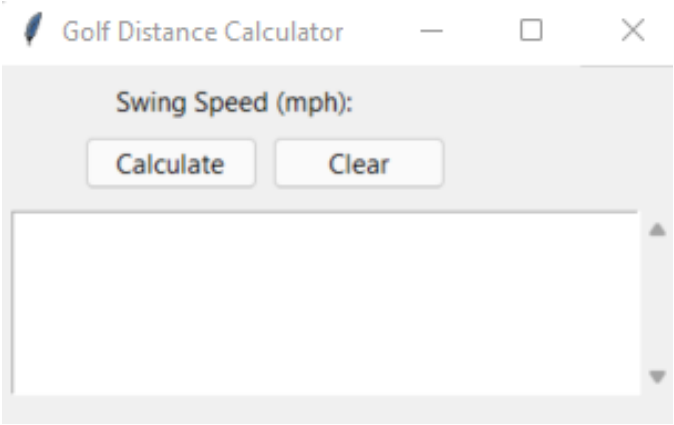


Figure Q4.2

(a) Illustrate your understanding of the given code in Table Q4 by showing the containment (parent-child) hierarchy of the GUI components.

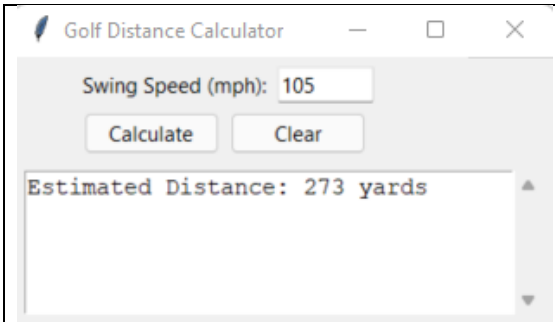
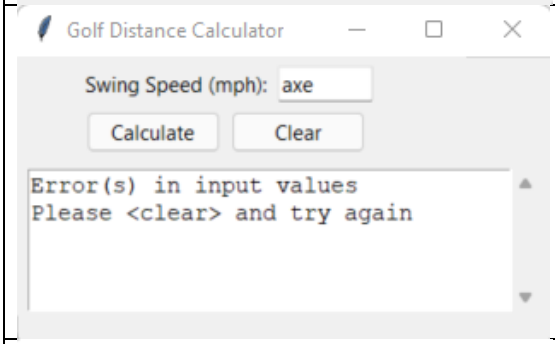
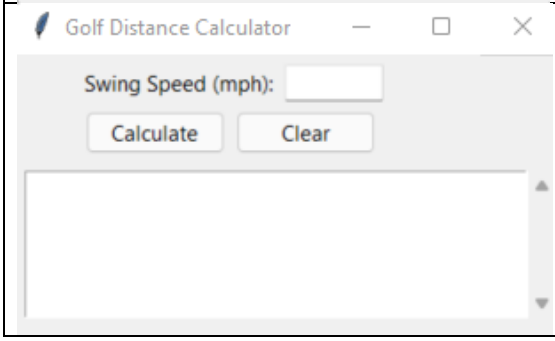
(8 marks)

(b) Modify and/or add code to the given code in Table Q4 to make the GUI appear as shown in Figure Q4.1

(4 marks)

(c) Implement event handling so that the GUI can respond to left button clicks on both buttons.

Below are demonstrations of the Golf Distance Calculator:

	When user enters a swing speed and click the "Calculate" button, your program will compute the distance in yards based on the following formula: Distance = Swing Speed * 2.6 Round the computed distance to nearest whole number before displaying.
	Your program should display error messages in the scrolledText when the swing speed data entry contains: - empty or no data value - non-numeric values - negative numbers
	When "Clear" button is clicked, your program should: - clear the "Swing Speed" text field - clear the ScrolledText

(13 marks)

Submit your codes for Q4(a) separately from Q4(b).

----- END OF PAPER -----